

Literatur

- [Acc24] Java Native Access. *Java-Native-Access/Jna*. Java Native Access, 1. Jan. 2024. URL: <https://github.com/java-native-access/jna> (besucht am 01.01.2024).
- [Ado23] AdoptOpenJDK. *JITWatch*. AdoptOpenJDK, 16. Sep. 2023. URL: <https://github.com/AdoptOpenJDK/jitwatch> (besucht am 24.09.2023).
- [AV05] Anderson Faustino da Silva und Vitor Santos Costa. „An Experimental Evaluation of JAVA JIT Technology“. In: (2005). DOI: 10.3217/JUCS-011-07-1291. URL: <https://lib.jucs.org/article/28447/> (besucht am 23.09.2023).
- [Bal] Ionut Balosin. *Analyzing JVM Energy Consumption for JDK 17: An Empirical Study – Ionut Balosin*. URL: <https://ionutbalosin.com/2023/07/analyzing-jvm-energy-consumption-for-jdk-17-an-empirical-study/> (besucht am 04.01.2024).
- [Bal23] Ionut Balosin. *JVM Performance Comparison for JDK 17 – Ionut Balosin*. 19. Juni 2023. URL: <https://ionutbalosin.com/2023/03/jvm-performance-comparison-for-jdk-17/> (besucht am 08.10.2023).
- [Bla+06] Stephen M. Blackburn u. a. „The DaCapo Benchmarks: Java Benchmarking Development and Analysis“. In: *Proceedings of the 21st Annual ACM SIGPLAN Conference on Object-oriented Programming Systems, Languages, and Applications*. OOPSLA06: ACM SIGPLAN Object Oriented Programming Systems and Applications Conference. Portland Oregon USA: ACM, 16. Okt. 2006, S. 169–190. ISBN: 978-1-59593-348-5. DOI: 10.1145/1167473.1167488. URL: <https://dl.acm.org/doi/10.1145/1167473.1167488> (besucht am 04.01.2024).
- [BM94] Marc M. Brandis und Hanspeter Mössenböck. „Single-Pass Generation of Static Single-Assignment Form for Structured Languages“. In: *ACM Transactions on Programming Languages and Systems* 16.6 (1. Nov. 1994), S. 1684–1698. ISSN: 0164-0925. DOI: 10.1145/197320.197331. URL: <https://dl.acm.org/doi/10.1145/197320.197331> (besucht am 07.10.2023).
- [Boo] Spring Boot. *Spring Boot Documentation: GraalVM Native Image Support*. URL: <https://docs.spring.io/spring-boot/docs/current/reference/html/native-image.html> (besucht am 15.11.2023).
- [Boo23] Spring Boot. *Spring PetClinic*. Spring, 16. Dez. 2023. URL: <https://github.com/spring-projects/spring-petclinic> (besucht am 16.12.2023).

- [Com23] Commend. *Commend International*. Commend International. 21. Apr. 2023. URL: <https://www.commend.com/de/unternehmen/commend-international.html> (besucht am 01.01.2024).
- [CR20] John Chapin und Mike Roberts. *Programming AWS Lambda: Build and Deploy Serverless Applications with Java*. Ö'Reilly Media, Inc., 18. März 2020. 278 S. ISBN: 978-1-4920-4102-3. Google Books: HdPXDwAAQBAJ.
- [DaC] DaCapo. *DaCapo Benchmarks*. DaCapo Benchmarks. URL: <http://dacapobench.org/> (besucht am 05.01.2024).
- [Dub+13] Gilles Duboscq u. a. „Graal IR: An Extensible Declarative Intermediate Representation“. In: 23. Feb. 2013.
- [Fou] Apache Software Foundation. *Apache JMeter - Apache JMeter™*. URL: <https://jmeter.apache.org/> (besucht am 16.12.2023).
- [GBC12] Luca Gherardi, Davide Brugali und Daniele Comotti. „A Java vs. C++ Performance Evaluation: A 3D Modeling Benchmark“. In: *Lecture Notes in Computer Science*. Bd. 7628. 5. Nov. 2012. ISBN: 978-3-642-34326-1. DOI: 10.1007/978-3-642-34327-8_17.
- [Gri+15] Matthias Grimmer u. a. „High-Performance Cross-Language Interoperability in a Multi-Language Runtime“. In: *Proceedings of the 11th Symposium on Dynamic Languages*. DLS 2015. New York, NY, USA: Association for Computing Machinery, 21. Okt. 2015, S. 78–90. ISBN: 978-1-4503-3690-1. DOI: 10.1145/2816707.2816714. URL: <https://doi.org/10.1145/2816707.2816714> (besucht am 01.11.2023).
- [Hel] Helidon. *Helidon Project*. URL: <https://helidon.io> (besucht am 15.11.2023).
- [HNG14] Tobias Hartmann, Albert Noll und Thomas Gross. „Efficient Code Management for Dynamic Multi-Tiered Compilation Systems“. In: *Proceedings of the 2014 International Conference on Principles and Practices of Programming on the Java Platform: Virtual Machines, Languages, and Tools*. PPPJ '14. New York, NY, USA: Association for Computing Machinery, 23. Sep. 2014, S. 51–62. ISBN: 978-1-4503-2926-2. DOI: 10.1145/2647508.2647513. URL: <https://doi.org/10.1145/2647508.2647513> (besucht am 23.09.2023).
- [Jeta] JetBrains. *FAQ / Kotlin*. Kotlin Help. URL: <https://kotlinlang.org/docs/faq.html> (besucht am 05.01.2024).
- [Jetb] JetBrains. *Kotlin 1.9.0: 'kotlin.Annotation.AnnotationTarget' Was Unintentionally Initialized at Build Time (GraalVM 'nativeTest') : KT-60211*. URL: <https://youtrack.jetbrains.com/issue/KT-60211> (besucht am 01.01.2024).
- [Kto] Ktor. *GraalVM / Ktor*. Ktor Help. URL: <https://ktor.io/docs/2.3.7/graalvm.html> (besucht am 01.01.2024).

- [Kum21] A. B. Vijay Kumar. *Supercharge Your Applications with GraalVM: Hands-on Examples to Optimize and Extend Your Code Using GraalVM's High Performance and Polyglot Capabilities*. Packt Publishing Ltd, 10. Aug. 2021. 360 S. ISBN: 978-1-80056-623-1. Google Books: j3wwEAAAQBAJ.
- [Lat] Chris Lattner. *The Architecture of Open Source Applications (Volume 1)LLVM*. URL: <https://aosabook.org/en/v1/llvm.html> (besucht am 05.01.2024).
- [Lin] Johannes Kepler Universität Linz. *Compiler and JVM Research at JKU*. URL: <https://ssw.jku.at/Research/Projects/JVM/> (besucht am 07.10.2023).
- [LLV] LLVM. *The LLVM Compiler Infrastructure Project*. URL: <https://llvm.org/> (besucht am 05.01.2024).
- [Log] Logback. *Logback: Setup*. URL: <https://logback.qos.ch/setup.html> (besucht am 04.01.2024).
- [M23] David M. Hypfvieh/Dbus-Java. 8. Dez. 2023. URL: <https://github.com/hypfvieh/dbus-java> (besucht am 01.01.2024).
- [Mica] MicroDoc. *Graal VM TM*. MicroDoc. URL: <https://www.microdoc.com/graal-vm/> (besucht am 04.01.2024).
- [Micb] Micronaut. *Micronaut Framework*. Micronaut Framework. URL: <https://micronaut.io/> (besucht am 15.11.2023).
- [Opea] OpenJDK. *JEP 335: Deprecate the Nashorn JavaScript Engine*. URL: <https://openjdk.org/jeps/335> (besucht am 01.01.2024).
- [Opeb] OpenJDK. *JEP 372: Remove the Nashorn JavaScript Engine*. URL: <https://openjdk.org/jeps/372> (besucht am 01.01.2024).
- [Opec] OpenJDK. *JEP 403: Strongly Encapsulate JDK Internals*. URL: <https://openjdk.org/jeps/403> (besucht am 01.01.2024).
- [Oped] OpenJDK. *Project Leyden: Beginnings*. URL: <https://openjdk.org/projects/leyden/notes/01-beginnings> (besucht am 19.01.2024).
- [Ope23a] OpenJDK. *Code Tools: JMH JDK Microbenchmarks*. OpenJDK, 4. Okt. 2023. URL: <https://github.com/openjdk/jmh-jdk-microbenchmarks> (besucht am 15.10.2023).
- [Ope23b] OpenJDK. *Java Microbenchmark Harness (JMH)*. OpenJDK, 14. Okt. 2023. URL: <https://github.com/openjdk/jmh> (besucht am 14.10.2023).
- [Oraa] Oracle. *Accessing Resources in Native Image*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/dynamic-features/Resources/> (besucht am 01.01.2024).
- [Orab] Oracle. *Graaljs/Docs/User/NashornMigrationGuide.Md at Master · Oracle/-Graaljs*. GitHub. URL: <https://github.com/oracle/graaljs/blob/master/docs/user/NashornMigrationGuide.md> (besucht am 04.11.2023).

- [Orac] Oracle. *GraalVM Documentation: Academic Publications*. URL: <https://www.graalvm.org/latest/community/publications/> (besucht am 01.11.2023).
- [Orad] Oracle. *GraalVM Documentation: Advantages for Java*. URL: <https://www.graalvm.org/java/advantages/> (besucht am 20.09.2023).
- [Orae] Oracle. *GraalVM Documentation: Class Initialization in Native Image*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/optimizations-and-performance/ClassInitialization/> (besucht am 21.10.2023).
- [Oraf] Oracle. *GraalVM Documentation: Graal Compiler*. URL: <https://www.graalvm.org/latest/reference-manual/java/compiler/> (besucht am 04.10.2023).
- [Orag] Oracle. *GraalVM Documentation: GraalVM LLVM Runtime*. URL: <https://www.graalvm.org/latest/reference-manual/llvm/> (besucht am 11.11.2023).
- [Orah] Oracle. *GraalVM Documentation: GraalVM Python Runtime*. URL: <https://www.graalvm.org/latest/reference-manual/python/> (besucht am 11.11.2023).
- [Orai] Oracle. *GraalVM Documentation: Interaction of GraalVM with Native Code*. URL: <https://www.graalvm.org/latest/reference-manual/llvm/NativeExecution/> (besucht am 11.11.2023).
- [Oraj] Oracle. *GraalVM Documentation: Java on Truffle*. URL: <https://www.graalvm.org/latest/reference-manual/java-on-truffle/> (besucht am 02.11.2023).
- [Orak] Oracle. *GraalVM Documentation: Memory Management*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/optimizations-and-performance/MemoryManagement/> (besucht am 22.10.2023).
- [Oral] Oracle. *GraalVM Documentation: Native Image*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/> (besucht am 20.09.2023).
- [Oram] Oracle. *GraalVM Documentation: Native Image Build Output*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/overview/BuildOutput/> (besucht am 22.10.2023).
- [Oran] Oracle. *GraalVM Documentation: Native Image Compatibility Guide*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/metadata/Compatibility/> (besucht am 21.10.2023).
- [Orao] Oracle. *GraalVM Documentation: On-Stack Replacement (OSR)*. URL: <https://www.graalvm.org/latest/graalvm-as-a-platform/language-implementation-framework/OnStackReplacement/> (besucht am 02.11.2023).

- [Orap] Oracle. *GraalVM Documentation: Polyglot Programming*. URL: <https://www.graalvm.org/latest/reference-manual/polyglot-programming/> (besucht am 20.09.2023).
- [Oraq] Oracle. *GraalVM Documentation: Static and Mostly Static Images*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/StaticImages/> (besucht am 15.11.2023).
- [Orar] Oracle. *GraalVM Documentation: Truffle Native Function Interface*. URL: <https://www.graalvm.org/latest/graalvm-as-a-platform/language-implementation-framework/NFI/> (besucht am 02.11.2023).
- [Oras] Oracle. *GraalVM on ARMv7 32 Bits · Issue #1329 · Oracle/Graal*. GitHub. URL: <https://github.com/oracle/graal/issues/1329> (besucht am 01.01.2024).
- [Orat] Oracle. *GraalVM Release Notes: GraalVM for JDK 21*. URL: https://www.graalvm.org/release-notes/JDK_21/ (besucht am 04.10.2023).
- [Orau] Oracle. *GraalVM Website: Presentation Materials*. URL: <https://www.graalvm.org/community/assets/> (besucht am 04.11.2023).
- [Orav] Oracle. *GuardNode (Gaal OpenJDK Project Documentation)*. URL: <http://lafo.ssw.uni-linz.ac.at/javadoc/graalvm/all/com/oracle/graal/nodes/GuardNode.html> (besucht am 14.10.2023).
- [Oraw] Oracle. *Ideal Graph Visualizer*. URL: <https://www.graalvm.org/latest/tools/igv/> (besucht am 07.10.2023).
- [Orax] Oracle. *JDK 9 Release Notes*. URL: <https://www.oracle.com/java/technologies/javase/9-relnotes.html> (besucht am 01.01.2024).
- [Oray] Oracle. *Libraries and Frameworks Ready for GraalVM Native Image*. URL: <https://www.graalvm.org/native-image/libraries-and-frameworks/> (besucht am 15.11.2023).
- [Oraz] Oracle. *Migration Guide from Nashorn to GraalVM JavaScript*. URL: <https://www.graalvm.org/latest/reference-manual/js/NashornMigrationGuide/> (besucht am 01.01.2024).
- [Oraaa] Oracle. *Native Image Compatibility Guide*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/metadata/Compatibility/> (besucht am 01.01.2024).
- [Oraab] Oracle. *Optimize Cloud Native Java Applications with GraalVM Enterprise PGO*. Oracle Help Center. URL: <https://docs.oracle.com/en/learn/build-optimized-cloud-native-java-applications-with-graalvm-enterprise-pgo/index.html#introduction> (besucht am 26.10.2023).
- [Oraac] Oracle. *Oracle Documentation: Collect Metadata with the Tracing Agent*. URL: <https://www.graalvm.org/latest/reference-manual/native-image/metadata/AutomaticMetadataCollection/> (besucht am 26.10.2023).

- [Oraad] Oracle. *Oracle GraalVM Enterprise Edition 20 Documentation*. Oracle Help Center. URL: <https://docs.oracle.com/en/graalvm/enterprise/20/docs/reference-manual/native-image/Reflection/#reflection-use-in-native-images> (besucht am 22.10.2023).
- [Oraae] Oracle. *The Java HotSpot Performance Engine Architecture*. URL: <https://www.oracle.com/java/technologies/whitepaper.html> (besucht am 23.09.2023).
- [Oraaf] Oracle. *Using Java Reflection*. URL: <https://www.oracle.com/technical-resources/articles/java/javareflection.html> (besucht am 21.10.2023).
- [Oraag] Oracle. *Using JNA inside a Native Image Does Not Work · Issue #673 · Oracle/Graal*. URL: <https://github.com/oracle/graal/issues/673> (besucht am 01.01.2024).
- [Oraah] Oracle. *Was ist GraalVM*. URL: <https://www.oracle.com/de/java/graalvm/what-is-graalvm/> (besucht am 04.10.2023).
- [Ora22] Oracle. *OpenJDK Announcement*. 2022. URL: <https://www.graalvm.org/2022/openjdk-announcement/> (besucht am 04.10.2023).
- [Ora23a] Oracle. *GraalVM Reachability Metadata Repository*. Oracle, 19. Okt. 2023. URL: <https://github.com/oracle/graalvm-reachability-metadata> (besucht am 21.10.2023).
- [Ora23b] Oracle. *Graalvm/Mx*. GraalVM, 27. Dez. 2023. URL: <https://github.com/graalvm/mx> (besucht am 04.01.2024).
- [Ora23c] Oracle. *Oracle/Graaljs*. Oracle, 3. Nov. 2023. URL: <https://github.com/oracle/graaljs> (besucht am 04.11.2023).
- [Ora24] Oracle. *Oracle/Graal*. Oracle, 4. Jan. 2024. URL: <https://github.com/oracle/graal> (besucht am 04.01.2024).
- [Pro+19] Aleksandar Prokopec u. a. „Renaissance: A Modern Benchmark Suite for Parallel Applications on the JVM“. In: *Proceedings Companion of the 2019 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity*. SPLASH Companion 2019. New York, NY, USA: Association for Computing Machinery, 20. Okt. 2019, S. 11–12. ISBN: 978-1-4503-6992-3. DOI: 10.1145/3359061.3362778. URL: <https://doi.org/10.1145/3359061.3362778> (besucht am 04.01.2024).
- [Qua] Quarkus. *Quarkus - Supersonic Subatomic Java*. URL: <https://quarkus.io/> (besucht am 15.11.2023).

- [RGM16] Manuel Rigger, Matthias Grimmer und Hanspeter Mössenböck. „Sulong - Execution of LLVM-based Languages on the JVM: Position Paper“. In: *Proceedings of the 11th Workshop on Implementation, Compilation, Optimization of Object-Oriented Languages, Programs and Systems*. ICPOOLPS '16. New York, NY, USA: Association for Computing Machinery, 17. Juli 2016, S. 1–4. ISBN: 978-1-4503-4837-9. DOI: 10.1145/3012408.3012416. URL: <https://doi.org/10.1145/3012408.3012416> (besucht am 15.11.2023).
- [Ros14] John Rose. *JEP 243: Java-Level JVM Compiler Interface*. 29. Okt. 2014. URL: <https://openjdk.org/jeps/243> (besucht am 04.10.2023).
- [Sch20] Roland Schatz. *Safe and Sandboxed Execution of Native Code*. graalvm. 9. Juni 2020. URL: <https://medium.com/graalvm/safe-and-sandboxed-execution-of-native-code-f6096b35c360> (besucht am 11.11.2023).
- [SWM14] Lukas Stadler, Thomas Wuerthinger und Hanspeter Mössenböck. „Partial Escape Analysis and Scalar Replacement for Java“. In: *Proceedings of the 12th ACM/IEEE International Symposium on Code Generation and Optimization, CGO 2014*. 15. Feb. 2014. DOI: 10.1145/2544137.2544157.
- [Vai23] Lionel Sujay Vailshery. *Most Used Languages among Software Developers Globally 2023*. Statista. 2023. URL: <https://www.statista.com/statistics/793628/worldwide-developer-survey-most-used-languages/> (besucht am 20.09.2023).
- [Wim+19] Christian Wimmer u. a. „Initialize Once, Start Fast: Application Initialization at Build Time“. In: *Proceedings of the ACM on Programming Languages* 3 (OOPSLA 10. Okt. 2019), 184:1–184:29. DOI: 10.1145/3360610. URL: <https://dl.acm.org/doi/10.1145/3360610> (besucht am 21.10.2023).
- [Wim17] Christian Wimmer. *Graal: High Performance Compilation for Managed Languages (PLDI 2017 - PLDI Tutorials) - PLDI 2017*. 23. Juni 2017. URL: <https://pldi17.sigplan.org/details/pldi-2017-workshops-and-tutorials/10/Graal-High-Performance-Compilation-for-Managed-Languages> (besucht am 07.10.2023).
- [Wim21] Christian Wimmer. „GraalVM Native Image: Large-Scale Static Analysis for Java (Keynote)“. In: *Proceedings of the 13th ACM SIGPLAN International Workshop on Virtual Machines and Intermediate Languages*. VMIL 2021. New York, NY, USA: Association for Computing Machinery, 19. Okt. 2021, S. 3. ISBN: 978-1-4503-9109-2. DOI: 10.1145/3486606.3488075. URL: <https://doi.org/10.1145/3486606.3488075> (besucht am 20.09.2023).
- [Wür+13] Thomas Würthinger u. a. „One VM to Rule Them All“. In: *Proceedings of the 2013 ACM International Symposium on New Ideas, New Paradigms, and Reflections on Programming & Software*. Onward! 2013. New York, NY, USA: Association for Computing Machinery, 29. Okt. 2013, S. 187–204. ISBN: 978-1-4503-2472-4. DOI: 10.1145/2509578.2509581. URL: <https://doi.org/10.1145/2509578.2509581> (besucht am 01.11.2023).

- [Wür+17] Thomas Würthinger u. a. „Practical Partial Evaluation for High-Performance Dynamic Language Runtimes“. In: *Proceedings of the 38th ACM SIGPLAN Conference on Programming Language Design and Implementation*. PLDI 2017. New York, NY, USA: Association for Computing Machinery, 14. Juni 2017, S. 662–676. ISBN: 978-1-4503-4988-8. DOI: 10 . 1145 / 3062341 . 3062381. URL: <https://doi.org/10.1145/3062341.3062381> (besucht am 02.11.2023).